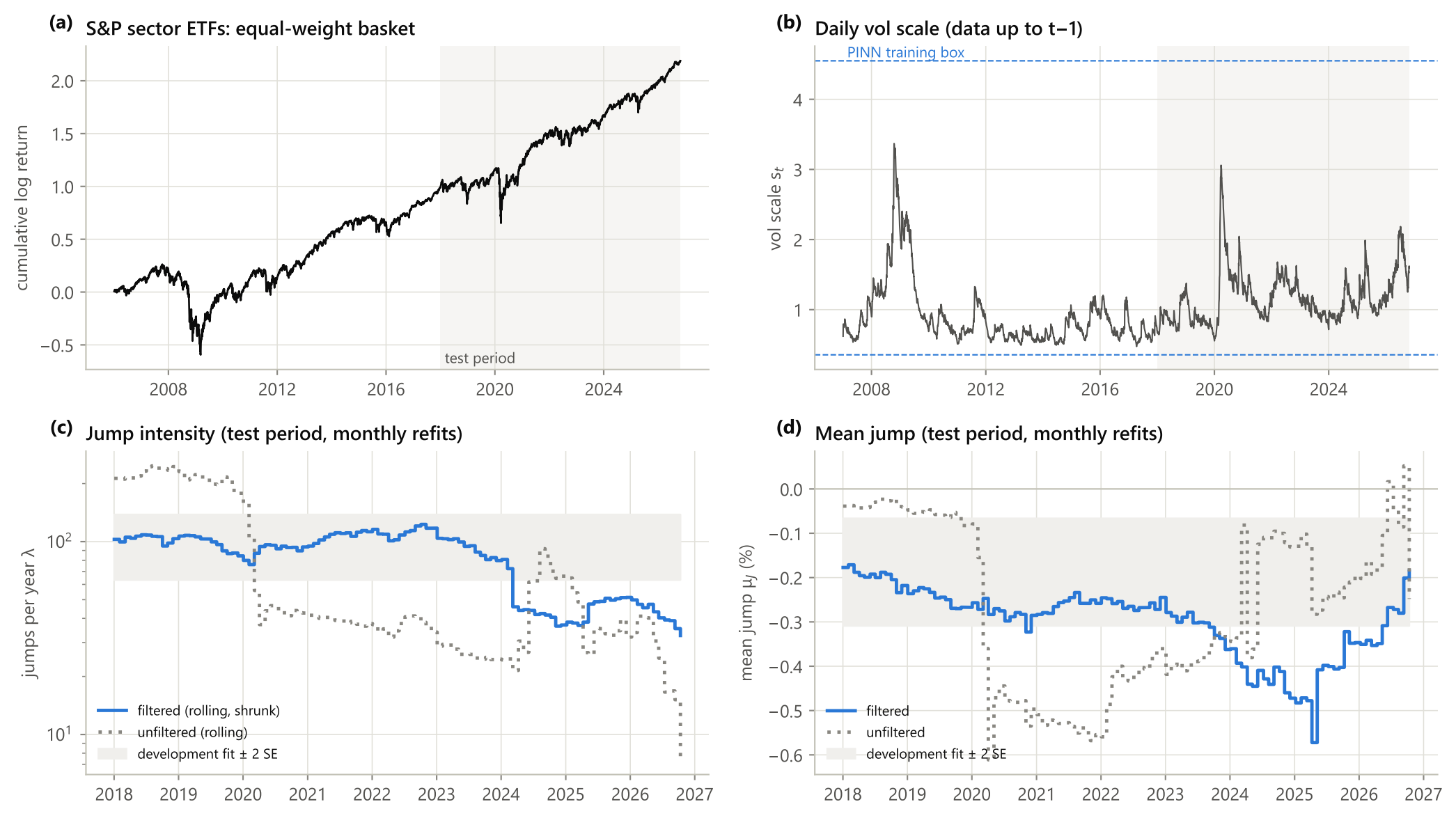

(a)
S&P sector ETFs: equal-weight basket
cumulative log return
2.0
1.5
1.0
0.5
0.0
−0.5
test period
2008
2012
2016
2020
2024
(b)
Daily vol scale (data up to t−1)
PINN training box
vol scale s_t
4
3
2
1
2008
2012
2016
2020
2024
(c)
Jump intensity (test period, monthly refits)
jumps per year λ
10^2
10^1
filtered (rolling, shrunk)
unfiltered (rolling)
development fit ± 2 SE
2018
2019
2020
2021
2022
2023
2024
2025
2026
2027
(d)
Mean jump (test period, monthly refits)
mean jump μ_J (%)
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
−0.6
filtered
unfiltered
development fit ± 2 SE
2018
2019
2020
2021
2022
2023
2024
2025
2026
2027